TELESCOPES FOR BEGINNERS

Choosing a telescope to suit your needs and experience can make a difference to your viewing. Many people start with a basic telescope, possibly on an altazimuth mount (see p.83), such as a Dobsonian, then learn how to find objects in the sky. Others opt for a more advanced computerized go-to telescope that will find celestial objects at the press of a button. These are invaluable for locating hard-to-find objects, and most include a sky-tour that will show the highlights visible at any given time and provide background information. Once an object has been located, the instrument will track it automatically for as long as required. Go-to telescopes may have altazimuth or equatorial mounts—the former are fine for visual observing, the latter are essential for long-exposure photography. Consider, too, whether you need your telescope to be portable. In general, the larger the telescope's aperture the better, but there is no point in having an instrument that you rarely use because it is too big and cumbersome to set up. Most instruments perform well on all subjects. Refracting telescopes tend to be better suited to use in towns, where light pollution can be a problem, while country sites favor reflecting telescopes.

FINDERS

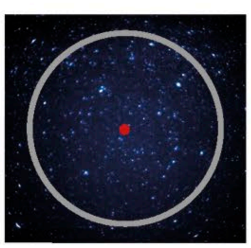
A telescope's field of view is small, even if using the lowest magnification eyepiece. It is typically only 1°, which is just twice the size of the full Moon in the sky. Simply aiming the telescope can be hit-or-miss. A finder, which is a small refracting telescope that sits on the side of the main instrument, helps you aim your telescope with much greater precision. Almost all telescopes require a finder to help locate objects, or for go-to telescopes to set them up in the first place. There are two main types: finderscopes and red dot finders. Optical finderscopes are useful where there is light pollution because they can show stars not otherwise visible, although red dot finders can be easier to use. Both are mounted on the telescope tube in such a way that they can be adjusted to match the aiming point of the main instrument (see below right). Align the finder by looking at a distant fixed object, ideally in daylight. Never use the Sun, which could blind you. Switch the finder off after using it to avoid a dead battery later.

GO-TO TELESCOPE
This equatorially mounted Schmidt–Cassegrain telescope is a typical computerized instrument. It has a handset for entering the details of target objects and a hand-held controller for adjustments in right ascension and declination. It can also interface with a computer.

FINDERSCOPE
A finderscope magnifies the night sky and gives a field of view of around 5–8°. A crosshair helps center the target in the finderscope. The image through the finderscope is inverted, which can, at first, make finding objects frustrating. Most entry-level telescopes come with a basic finder, but it may be worth upgrading as you progress.

FINDERSCOPE VIEW

RED DOT FINDER
A red dot finder indicates where the telescope is pointing by projecting a small red dot onto a piece of transparent glass or plastic. The wider sky remains visible, making it intuitive to use. The brightness of the dot can sometimes be adjusted with a built-in dimmer switch.



RED DOT FINDER VIEW

USING A FINDER

1 FIND AN OBJECT
To align any finder, first select a distant and fixed object in the main instrument using its lowest magnification eyepiece, then center it within the field of view. Clamp the telescope's position.

2 ALIGN THE FINDER
Using the adjusters on the finder, bring the same object into the center of the finder's crosshairs or red dot (see far left). You may need to repeat this each time the finder is removed and replaced.